Math 1452 Calculus II

Professor: Dr. Brock Williams	**Place:** Math 109
Office Hours: 11-12 MW, and by appointment	**Text:** *Calculus* by Smith/Strauss/Toda
Office: Math 244	**Time:** 1-2:50 PM MW
Phone: 742-2566	**Prerequisites:** Math 1451 or equivalent
Email: brock.williams@ttu.edu	**Final Exam:** Monday, May 11, 10:30AM -1:00PM

TTU Core Curriculum Student Learning Outcomes for Mathematics

- Apply arithmetic, algebra, geometry and statistics to solve problems.
- Represent and evaluate basic mathematical information numerically, graphically, and symbolically.
- Use mathematical and logical reasoning to evaluate the validity of an argument.
- Interpret mathematical models such as formulas, graphs, tables and schematics, and draw inference from them.

Students graduating from Texas Tech University should be able to demonstrate the ability to apply quantitative and logical skills to solve problems. This course satisfies core curriculum math requirements.

Expected Learning Outcomes	Methods for Assessment
Set up and evaluate integrals to find areas and volumes and to solve real world problems	Exam I, Homework, Class Discussion, Final Exam
Evaluate integrals by hand using a variety of techniques including substitutions, parts, partial fractions, and hyperbolic trigonometry	Exam II, Homework, Class Discussion, Final Exam
Analyze the convergence of sequences, series, and power series	Exam III, Homework, Class Discussion, Final Exam
Solve elementary problems in vector analysis	Homework, Class Discussion, Final Exam

Major Course Requirements: There will be 3 in-class exams, each accounting for 20% of your course grade. Another 15% of the grade will be derived from the homework, quizzes, in-class activities, projects, etc. The remaining 25% will consist of your score on the comprehensive departmental final exam. The grading scale is

Percentage	Grade
90-100	A
80-89	B
70-79	C
60-69	D
Below 60	F

Some minor downward adjustments may be made at the end of the semester; however, any such adjustments will be solely at the discretion of the instructor. Under no circumstances should you count on this possibility. No "curves" for exam scores should be expected.

Schedule	
Lecture	**General Description of the Subject Matter**
Week 1-5	Using integrals to find areas and volumes and to solve real world problems
Week 6-9	Evaluating integrals by hand
Week 10-14	Analyzing the convergence of sequences, series, and power series
Week 15-16	Vector analysis and review

Required Reading: You are expected to read the relevant sections of the textbook before attending each lecture.

Attendance: You are responsible for all material covered and announcements made in lecture, via email, and on the web. It is critical that you come to every class. Attendance will be recorded. Your final average will be adjusted by 0.5 points * (3 - number of absences from class). For example, if you have 4 absences, your final average will be lowered by 0.5 points. On the other hand, if you do not miss any class, your average will be raised by 1.5 points.

Online Lectures: Lectures from previous semesters have been posted at mediacast.ttu.edu. Log in with your eraider name and browse the catalog on the left (Mathematics & Statistics, then Courses). These can be very useful in reviewing the course material, but are no substitute for attending class.

Homework: Homework will be of two types: WebWork (completed online) and problems assigned from the textbook (turned in on paper). WebWork homework will count twice as much as the paper homework. You are encouraged to work with other students; however, your work must be essentially your own.

Textbook: You are required to have a copy of the textbook, but are free to purchase it from any seller you choose. You are not required to purchase any additional book or notes.

Calculators: Use of calculators, phone apps, etc, is allowed on homework, but forbidden on all exams, including the final. Except for simple arithmetic and trig functions, you should not need a calculator even on the homework, and would be well-advised to avoid it as much as possible.

Academic Integrity: You are expected to maintain the highest standards of academic integrity in this course. Any violation will be punished to the fullest extent. In particular, no calculators, notes, books, or other outside resources may be used on exams. You may (and are encouraged to) discuss homework problems with each other and with tutors at the math tutoring center, but you may not use any work done completely by another person or copy from the student solution guide. For example, suppose you are working with a friend or tutor on a homework problem and they discover the answer. You may not copy their answer or their work. If however, they explain the problem to you, and you **completely** understand every detail (meaning you could work a similar problem completely on your own), then you may re-write the solution as you understand it and the solution would now be considered your own.

General Policies:
In general, no missed in-class activities will be made up and no homework will be accepted after the deadline. If you miss one of the 3 exams, your grade on the final can be used to replace one missed exam, but only with my permission for a valid, documented emergency.

Any student who, because of a disability, may require special arrangements in order to meet the course requirements should contact the instructor as soon as possible to make any necessary arrangements. Students should present appropriate verification from Student Disability Services during the instructor's office hours. Please note instructors are not allowed to provide classroom accommodations to a student until appropriate verification from Student Disability Services has been provided. For additional information, you may contact the Student Disability Services office in 335 West Hall or 806-742-2405.

Texas House Bill 256 requires institutions of higher education to excuse a student from attending classes or other required activities, including examinations, for the observance of a religious holy day. A student who intends to observe a religious holy day should make that intention known to the instructor prior to the absence. A student who is absent from classes for the observance of a religious holy day shall be allowed to take an examination or complete an assignment scheduled for that day within a reasonable time after the absence. A student may not be penalized for the absence; however, the instructor may respond appropriately if the student fails to complete the assignment satisfactorily.